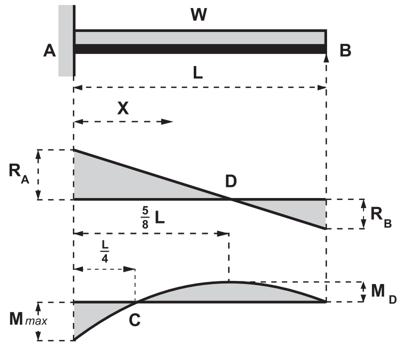
Propped cantilever — Uniform load on full span



Span $= L$

Total uniform load $= W$

$R_A = \frac{5}{8}W$ $R_B = \frac{3}{8}W$

at A $M_{max} = -\frac{WL}{8}$

at $\frac{5}{8}$L fromA $M_D = \frac{9}{128}WL$

at 0.5785L from A $\delta_{max} = \frac{WL^3}{185EI}$

at B $i_B = \frac{WL^2}{48EI}$

at X from A:

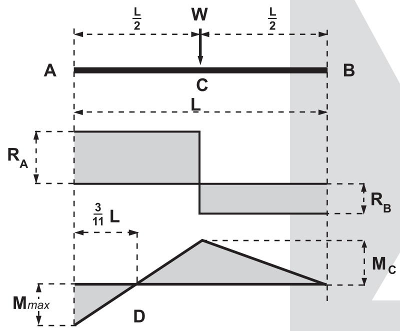
$M_X = \frac{W}{8L}(L^2\text{-}5LX+4X^2)$

$\delta_X = \frac{WX^2}{48EIL}(3L^2\text{-}5LX+2X^2)$

$i_x = \frac{WX}{48EIL}(6L^2\text{-}15LX+8X^2)$

at $\frac{1}{4}$L from A, $M_C = 0$

Propped cantilever — Point load at mid-span



Span $= L$

Point load $= W$

$R_A = \frac{11}{16}W$ $R_B = \frac{5}{16}W$

at A $M_{max} = -\frac{3}{16}WL$

at mid-span under load:

$M_C = \frac{5}{32}WL$

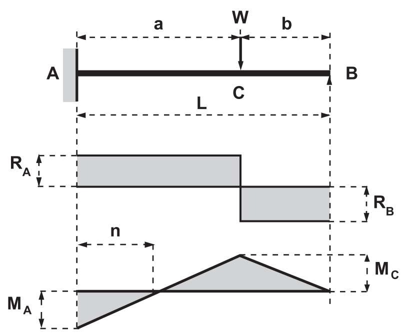
$\delta_C = \frac{7WL^3}{768EI}$

at 0.5528L from A $\delta_{max} = \frac{WL^3}{107EI}$

at B $i_B = \frac{WL^2}{32EI}$

at $\frac{3}{11}$L from A, $M = 0$

Propped cantilever — Point load at any position



Span $= L$

Point load $= W$

$R_A = \frac{Wb(3L^2\text{-}\ b^2)}{2L^3}$ $R_B = \frac{Wa^2(2L+b)}{2L^3}$

$M_A = -\frac{Wab(L+b)}{2L^2}$ $M_C = \frac{Wa^2b(2L+b)}{2L^3}$

$i_B = \frac{Wa^2b}{4EIL}$

Absolute max deflection is under the load when a=b√2=0.5858L $\delta_{max\ max} = \frac{WL^3}{102EI}$

When a>b√2 max deflection is between A and C $\delta_{max} = \frac{Wa^3b}{3EI} \cdot \frac{(L+b)^3}{(3L^2-b^2)^2}$

When a<b√2 max deflection is between C and B $\delta_{max} = \frac{Wa^2b}{6EI}\sqrt{\frac{b}{2L+b}}$

at n=aL $\frac{L+b}{3L^2\text{-}\ b^2}$ from A,M = 0